

Effects of Particle Concentration on the Partitioning of Suspensions at Small Divergent Bifurcations

R. Ditchfield

W. L. Olbricht

School of Chemical Engineering,
Cornell University,
Ithaca, NY 14853

Experimental results are reported for the low Reynolds number flow of a suspension of spherical particles through a divergent capillary bifurcation consisting of a straight tube of circular cross-section that splits to form two tubes of equal diameter. The partitioning of particles between the downstream branches of the bifurcation is measured as a function of the partitioning of total volume (particles + suspending fluid) between the branches. Two bifurcation geometries are examined: a symmetric Y-shaped bifurcation and a nonsymmetric T-shaped bifurcation. This experiment focuses on the role of hydrodynamic interactions between particles on the partitioning of particles at the bifurcation. The particle diameter, made dimensionless with respect to the diameter of the branch tubes, ranges from 0.4 to 0.8. Results show that hydrodynamic interactions among the particles are significant at the bifurcation, even for conditions where interactions are unimportant in the straight branches away from the bifurcation. As a result of hydrodynamic interactions among particles at the bifurcation, the partitioning of particles between the branches is affected for particle volume fractions as small as 2 percent. The experimental results show that the effect of particle volume fraction is to diminish the inhomogeneity of particle partitioning at the bifurcation. However, the magnitude of this effect depends strongly on the overall shape of the bifurcation geometry, and, in particular on the angles between the branches.

Introduction

An accurate description of the distribution of blood cells in microcirculatory networks is an important step toward developing reliable models of blood gas delivery and nutrient transport in tissue. The distribution of cells in microcirculatory networks can be strongly inhomogeneous, not only among the vessels of a network, but also within individual vessels. One source of the inhomogeneity is the distribution of cells at divergent capillary bifurcations. Observations show that the partitioning of cells between the downstream branches of a divergent bifurcation differs from the partitioning of total volume (cells and plasma) between the branches. The difference between the fraction of cells entering a downstream branch and the fraction of blood entering the same branch can be substantial. Indeed, when the fraction of blood volume entering one of the branches becomes sufficiently large, all red cells that flow through the bifurcation enter that branch. The other branch receives no cells, even though there is some plasma flow into that branch. The inhomogeneous partitioning of cells at divergent bifurcations can lead to significant inhomogeneities in the cell distribution that propagate throughout the interconnected vessels of the network.

Although the details of the flow at a capillary bifurcation are complicated and difficult to analyze, two qualitative mechanisms have been identified that can help explain the inhomogeneous partitioning of cells at bifurcations. The first mechanism, called plasma skimming, is a consequence of the fact that the radial distribution of cells is nonuniform in the inlet vessel upstream of the bifurcation. The nonuniform radial distribution

is a consequence of several factors, including the exclusion of cells from a thin layer near the vessel wall and migration of cells across streamlines in flow. Regardless of its origin, the nonuniform radial distribution of cells entering the bifurcation produces a difference between the fraction of cells entering a downstream branch and the fraction of total volume entering that branch. The second mechanism, called cell screening, is a consequence of hydrodynamic interactions between the cells and the vessel walls. Owing to hydrodynamic interactions, the motion of the cells differs from that of the surrounding plasma, producing a difference between the cell flux and the volume flux entering a downstream branch. For bifurcations composed of large vessels, plasma skimming is more important than cell screening, but for capillary bifurcations where the cell size is on the same order as the lumen diameter, both mechanisms are important.

There have been many experimental measurements of red cell partitioning at divergent capillary bifurcations, both in vitro and in vivo. The goal of most of these studies is to determine how the cell flux into the downstream branches of a bifurcation depends on such characteristics as the angles between the downstream branches, branch diameters, and inlet vessel hematocrit [e.g., 2, 3, 4, 5, 7, 8, 10]. However, the results of these experiments sometimes are difficult to interpret because it is usually impossible to vary one parameter at a time in any single experiment. The results are not completely general because it is difficult to vary experimental parameters of interest over wide ranges of values in any single experiment. Therefore, theoretical studies and model experiments involving well defined flows and precisely characterized synthetic systems can help characterize the influence of parameters on particle motions at bifurcations, which, in turn, can help interpret results of in vivo and in vitro blood flow experiments.

Theoretical and numerical studies of the problem are limited thus far by the complexity of flow, which is fully three-dimen-

Contributed by the Bioengineering Division for publication in the JOURNAL OF BIOMECHANICAL ENGINEERING. Manuscript received by the Bioengineering Division September 1, 1994; revised manuscript received June 14, 1995. Associate Technical Editor: S. P. Suter.

sional [6, 9] and unsteady in a Lagrangian sense when the motion of the cells is taken into account explicitly. However, a two-dimensional analog has been examined by computing the trajectory of a rigid particle through a divergent bifurcation using boundary integral methods [1]. An important qualitative result of the calculations, which is confirmed in the present three-dimensional model experiment, is that the translational speed of a cell decreases as it moves through a divergent bifurcation. The minimum speed of the cell depends on the particular trajectory it takes through the bifurcation. The cell speed becomes very small if the cell enters the bifurcation close to the special trajectory that divides cells between the two branches. Even though the cell speed decreases in the bifurcation, blood flow into the bifurcation continues. This suggests the possibility that cells may come in close proximity to each other and interact hydrodynamically in the bifurcation, even though they were spaced sufficiently far apart in the inlet vessel so that they did not interact there. If this is the case, then interactions among cells may affect blood flow and cell partitioning at divergent bifurcations for surprisingly small values of the inlet vessel tube hematocrit. It would then be relevant to determine whether hydrodynamic interactions among the cells at a bifurcation enhance or diminish the inhomogeneity of cell partitioning at the bifurcation.

Model experiments to simulate cell motion through bifurcations have employed a variety of particles, including spheres [2] and flexible disks [4], as model cells. Bugliarello and Hsiao [2] studied the effect of particle volume fraction in the inlet vessel on the distribution of spheres at a bifurcation. They found that the flux of spheres entering each of the downstream branches of a divergent bifurcation is influenced by the inlet branch hematocrit. However, these results pertain to vessels that are considerably larger than capillaries, because the sphere diameter was only 10 percent of the tube diameter, and because the apparent Reynolds number was about 50. We are not aware of any experiments that address quantitatively the effects of hematocrit on cell flux at capillary bifurcations. At bifurcations composed of capillaries, the cells occupy a significant fraction of the vessel lumen and the Reynolds number is small. Furthermore, local values of tube hematocrit can be considerably smaller in capillaries than in larger vessels.

Pries et al. [8] examined the flux of red cells at 65 arteriolar bifurcations in the rat mesentery to determine effects of bifurcation geometry and hematocrit on the cell flux. The results showed that although the cell flux into a given downstream vessel can differ substantially from the volumetric flow rate entering the same branch, the difference decreases as the value of the hematocrit increases. To obtain this result, data for all 65 bifurcations were lumped together, even though the bifurcations differed in vessel diameters and bifurcation geometry. Therefore, it is difficult to identify the exact mechanism that explains the effect of hematocrit on cell partitioning or to predict whether the results pertain qualitatively to any particular bifurcation with a specific geometry. In the absence of such information, model studies involving precisely characterized systems may prove useful in interpreting data such as those reported in [8].

Experimental

Apparatus and Materials. The central part of the apparatus, which is similar to the design in [5], consists of a straight tube of circular cross section that splits into two tubes at a bifurcation. The parameters that specify the geometry of the bifurcation are shown in Fig. 1. The bifurcation was machined from a $10 \times 10 \times 1$ cm Lucite block. The lengths of the inlet vessel and the two downstream branches were 14 cm and 12 cm, respectively. The block was machined simply by drilling holes of appropriate size and orientation. No attempt was made to smooth the wall contours at the bifurcation. The machined block was mounted vertically on a stage so that a videocamera

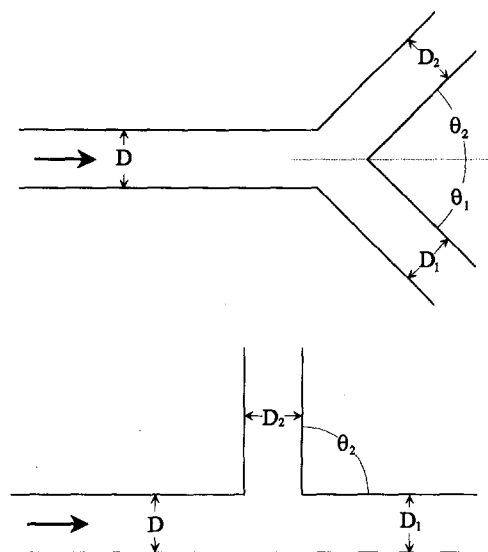


Fig. 1 Schematic showing parameters that specify the geometry of the bifurcation

mounted on an adjacent tripod could photograph the particles as they moved through the bifurcation. The images were recorded on videotape for analysis after the experiment.

To study the effect of branching angle on particle flux, we conducted studies for a Y-shaped bifurcation, i.e., with $\theta_1 = \theta_2 = \pi/4$, and a T-shaped bifurcation, i.e., with $\theta_1 = 0$ and $\theta_2 = \pi/2$. The branch diameters were identical for both bifurcations with $D = D_1 = D_2 = 0.8$ cm.

Figure 2 shows a schematic of the overall apparatus. The particles and suspending fluid were held in a 1 liter reservoir that was connected to the inlet vessel by a conical section. A pressure driven flow was generated by drawing a vacuum on the downstream branches. The flow rate in each branch was regulated by adjusting valves that connected the vacuum line to two 0.8 liter collection reservoirs downstream of the bifurcation. The volumetric flow rate in each branch was determined by measuring the time required to fill marked sections of the collection flasks.

The suspending fluid was polybutene (Indopol) which has a density of 0.875 g/cm^3 and a viscosity of 40 Poise. The particles were rigid polypropylene spheres (Precision Plastic Ball) with diameters of 0.32, 0.49, or 0.64 cm. The spheres were neutrally buoyant in the polybutene suspending fluid, and there was no evidence that the results were affected by the orientation of the flow cell with respect to gravity. Owing to the high viscosity of the polybutene, the Reynolds number based on the tube diameter was always smaller than 0.1.

Procedure. The dimensionless sphere size λ is defined as the sphere diameter divided by the tube diameter. The three sphere diameters gave λ values of 0.4, 0.6, and 0.8. For each value of λ , experiments were run for three particle volume fractions, ϕ , of 0.02, 0.08, and 0.15. Although these values may be smaller than microcirculatory hematocrits, the experiment is designed to explore the first effects of interactions among the particles as ϕ is increased from zero. To achieve the required volume fraction, 200 particles were mixed with the appropriate quantity of suspending fluid. Because the reservoir was connected to the inlet branch by a conical section, there were no appreciable entrance effects that could lead to particles being excluded at the entrance of the inlet branch. Therefore, the volume fraction ϕ is equivalent to a discharge hematocrit. However, the instantaneous value of the particle volume fraction in all branches of the bifurcation, the tube hematocrits, can differ from ϕ because the particle velocity is not identical to the average velocity of the flow.

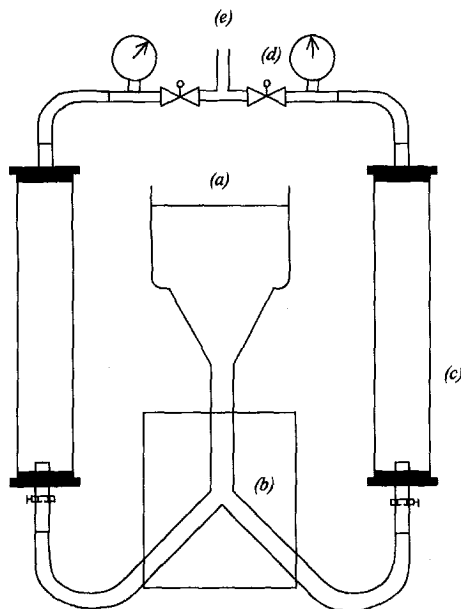


Fig. 2 Schematic of the experimental apparatus, including (a) a reservoir containing the suspension, (b) the machined bifurcation, (c) graduated cylinders into which the suspension is collected, (d) valves to control the flow rate in each branch of the bifurcation, and (e) connection to a vacuum pump

The value of the flow split, Q_1/Q_T , was set to its desired value by adjusting the valves connecting the downstream branches to the bifurcation and reading the pressures on two in-line gauges. Here, Q_1 is the volumetric flow rate into one of the branches and Q_T is the total volumetric flow rate entering the bifurcation. Because the Reynolds number is small, the absolute value of the flow rate has no consequence on the trajectory of the particles. Data collection did not begin until the bifurcation and the tubes that connect the bifurcation branches to the collection reservoirs were filled with fluid and some particles. The flow rate in each downstream branch was measured several times during the experiment by timing the rate at which the volumetric collection vessels filled. On rare occasions that the flow rate varied inadvertently during the experiment, the experiment was stopped and the procedure was repeated from the start.

The videotape of the bifurcation was reviewed after the experiment to count the number of particles that entered each branch. The result was used to calculate N_1/N_T , where N_1 is the number of particles entering one of the branches and N_T is the total number of particles that moved through the bifurcation. Individual frames of the videotape that show interesting features of the interactions between spheres at the bifurcation were digitized with a frame grabber board and software (Adobe Photoshop) and stored on computer disk (Apple IICx). Some of these frames are presented in Section 4.

Results

We consider first the results for the Y-shaped bifurcation. Following previous investigators, we present the fractional particle flux N_1/N_T entering one of the branches as a function of the fractional volumetric flow rate Q_1/Q_T entering that branch.

Figure 3 shows results for the largest particle size that was tested, $\lambda = 0.8$, for the three particle volume fractions, $\phi = 0.02, 0.08$, and 0.15 . Because the geometry of the Y-shaped bifurcation is symmetric, the data should be symmetric about the point $(0.5, 0.5)$. After verifying this was the case for several examples, we collected data only for $Q_1/Q_T \geq 0.5$ and completed the curves in the figures by reflecting results about the point $(0.5, 0.5)$.

For each particle volume fraction, when the flow split Q_1/Q_T is not too different from 0.5, the data fall on the diagonal line, which means the partitioning of particles is identical to the partitioning of total volume. However, when the value of the flow split becomes sufficiently large or sufficiently small, the data abruptly deviate from the line, and the particle flux favors the branch receiving the greater volumetric flow rate. The range of Q_1/Q_T values for which $N_1/N_T = Q_1/Q_T$ increases with increasing particle volume fraction ϕ . For $\phi = 0.15$, the data fall on the diagonal line over the range $0.26 < Q_1/Q_T < 0.74$. For $\phi = 0.08$, the range is $0.38 < Q_1/Q_T < 0.62$. For the most dilute system investigated, $\phi = 0.02$, the data fall on the diagonal only over a small range of about $0.46 < Q_1/Q_T < 0.54$. For all three particle volume fractions, the departure from the diagonal line is abrupt. Changes in Q_1/Q_T as small as 0.02 can yield significant deviations from the diagonal line at the limits of the ranges cited above.

For sufficiently large Q_1/Q_T , all the particles enter downstream branch 1, as expected. However, the value of Q_1/Q_T that is required for all the particles to enter branch 1 increases with increasing particle volume fraction ϕ .

Figure 4 shows the corresponding results for $\lambda = 0.6$ and $\lambda = 0.4$. In these cases, there is little difference between the partitioning of particles and the partitioning of total volume at the bifurcation, except when Q_1/Q_T is close to either unity or zero. For example, for $\lambda = 0.6$ and $\phi = 0.02$, there is no difference between particle flux and volumetric flow rate up to Q_1/Q_T of at least 0.78, but there is a difference for $Q_1/Q_T = 0.86$. For large and small values of Q_1/Q_T , the deviation from the line $N_1/N_T = Q_1/Q_T$ decreases as ϕ is made larger, as was the case for $\lambda = 0.8$. For $\lambda = 0.4$ there is very little deviation between particle flux and volumetric flow rate, except when Q_1/Q_T either exceeds 0.9 or is less than 0.1. Even then, the deviation is too small to make any comparisons among data for different values of ϕ .

We now turn to results for the T-shaped bifurcation. Because the downstream branches of the T-shaped bifurcation are not symmetric with respect to the inlet branch, it is important to specify the branch that will be used to present data. We arbitrarily use the straight branch as branch 1 and present data for N_1/N_T . Owing to the asymmetry of the bifurcation geometry, data must be collected over the full Q_1/Q_T range. The particle flux for the side branch N_2/N_T is simply $1 - N_1/N_T$.

Figure 5 shows the results for $\lambda = 0.8$ for the three volume fractions. The data show two features that are different from

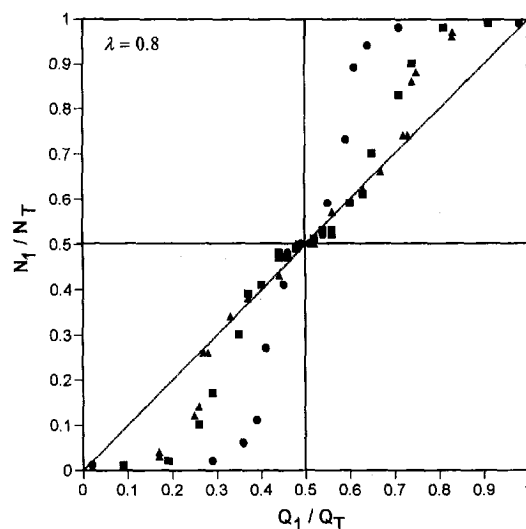


Fig. 3 The fractional cell flux entering branch 1 of a Y-shaped bifurcation for dimensionless particle size $\lambda = 0.8$ and for particle volume fractions $\phi = 0.02$ (●), 0.08 (■), and 0.15 (▲)

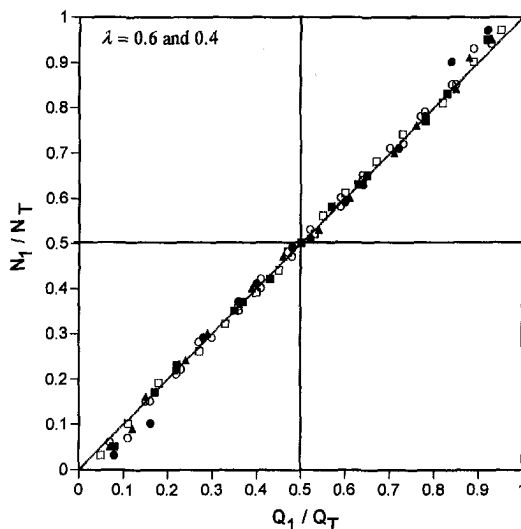


Fig. 4 The fractional cell flux entering branch 1 of a Y-shaped bifurcation for dimensionless particle size $\lambda = 0.6$, $\phi = 0.02$ (●), 0.08 (■), 0.15 (▲) and for $\lambda = 0.4$, $\phi = 0.02$ (○), 0.08 (□), 0.15 (△)

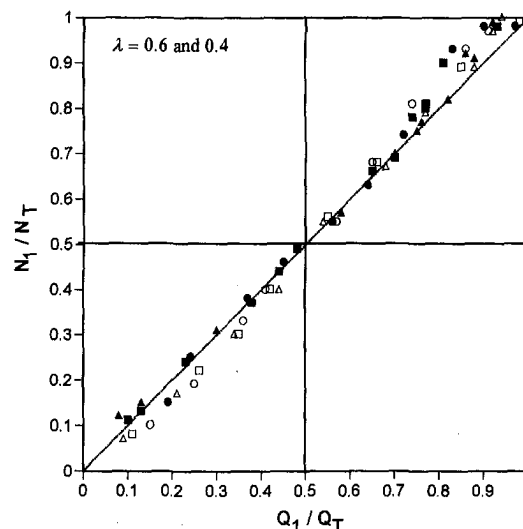


Fig. 6 The fractional cell flux entering branch 1 of a Y-shaped bifurcation for dimensionless particle size $\lambda = 0.6$, $\phi = 0.02$ (●), 0.08 (■), 0.15 (▲) and for $\lambda = 0.4$, $\phi = 0.02$ (○), 0.08 (□), 0.15 (△)

the results for the Y-shaped bifurcation. First, the values of N_1/N_T are not symmetric about the point (0.5, 0.5). Instead, N_1/N_T exceeds Q_1/Q_T by a substantial amount for $Q_1/Q_T \geq 0.5$, whereas N_1/N_T differs only slightly from Q_1/Q_T for $Q_1/Q_T < 0.5$. These results suggest that the particle flux "favors" the straight downstream branch over the side branch of the T-shaped bifurcation for nearly the entire range of Q_1/Q_T values. For Q_1/Q_T less than about 0.2, the value of N_1/N_T is smaller than Q_1/Q_T , but only for $\phi = 0.02$. If there are regions in which N_1/N_T is smaller than Q_1/Q_T for the other two values of ϕ , they must occur for values of Q_1/Q_T that are smaller than the minimum value covered in the experiment. Second, and perhaps more surprising, the data for the T-shaped bifurcation are relatively insensitive to the value of the particle volume fraction ϕ compared with the data for the symmetric Y-shaped bifurcation, especially when N_1/N_T deviates substantially from Q_1/Q_T for $Q_1/Q_T \geq 0.5$.

Figure 6 shows the flux of particles with $\lambda = 0.6$ and $\lambda = 0.4$ through the T-shaped bifurcation. For $\lambda = 0.6$ the value of N_1/N_T does not differ significantly from Q_1/Q_T over most of

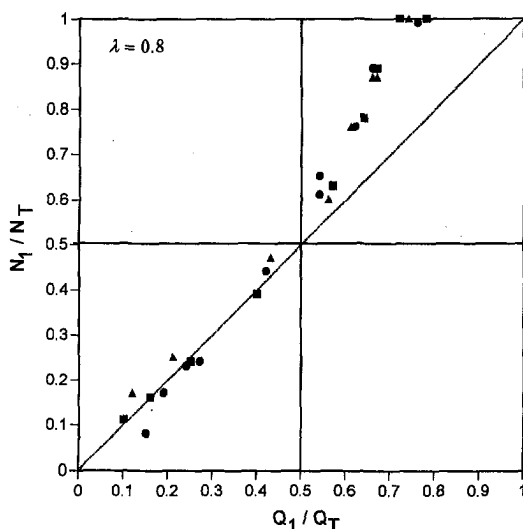


Fig. 5 The fractional cell flux entering branch 1 of a T-shaped bifurcation for dimensionless particle size $\lambda = 0.8$ and for particle volume fractions $\phi = 0.02$ (●), 0.08 (■), and 0.15 (▲)

the range of Q_1/Q_T values. Only when Q_1/Q_T exceeds about 0.75 is there a discernible difference between the particle flux and volumetric flux entering branch 1. In this case, as for the larger particles, the fractional particle flux entering the straight branch exceeds the fractional volumetric flux entering the branch. Again, the results are practically independent of the particle volume fraction. For $\lambda = 0.4$ the maximum difference between N_1/N_T and Q_1/Q_T is less than that for the larger particles. For $Q_1/Q_T > 0.5$ the results closely resemble those for the same particles in the Y-shaped bifurcation. However, for $Q_1/Q_T < 0.5$, the fractional particle flux entering the straight branch is less than the fractional volume flux entering the straight branch for all Q_1/Q_T .

Discussion

The results of this experiment show that the partitioning of particles at a divergent bifurcation depends on the partitioning of total volume, on the geometry of the bifurcation, and on the particle volume fraction. Moreover, the results show that the qualitative influence of each of these parameters on the particle flux depends on the values of the other parameters over much of the parameter ranges covered in the experiment. For example, results for the Y-shaped bifurcation show that the particle flux favors the branch receiving the greater volumetric flow rate, and the magnitude of this effect increases with particle size, which is in accord with previous experiments and numerical results [1, 4]. The new information in the present results is that the difference between the particle flux and volume flux into the branches of a Y-shaped bifurcation depends also on the particle volume fraction in the inlet branch. The effect of increasing particle concentration is to diminish the difference between the particle flux and the volume flux over a wide range of flow splits. However, the effect of particle concentration on particle partitioning clearly depends on the shape of the bifurcation, because results for the T-shaped bifurcation depend only weakly, if at all, on the particle volume fraction in the inlet branch.

The values of the particle volume fractions examined in this experiment are not large. For flow in a straight tube, interactions among the particles for values of ϕ between 0.02 and 0.08 are not likely to affect substantially the properties of the flow. However, the particle partitioning at a bifurcation is affected by interactions among the particles for small values of ϕ because the local number density of particles at the bifurcation is

larger than the value of ϕ suggests. The reason for this is that the translational velocity of a particle decreases as it moves through the bifurcation. Numerical results [1] show that a particle's translational velocity can become very small if the particle happens to be on a trajectory close to the special trajectory that divides particle trajectories between the downstream branches. In this case, there is sufficient time for the next particle to enter the bifurcation and interact with the slowly moving particle, even though these particles were initially spaced sufficiently far apart that they did not interact in the inlet vessel. Hydrodynamic interactions influence the particles' motions and the partitioning of the particles between the downstream branches, as can be inferred from the data. However, the mechanisms of interaction, and how they are affected by Q_1/Q_T and ϕ , can be discovered only by examining particle trajectories at the bifurcation from videotapes of the experiment.

Figure 7 shows a sequence of images for spheres with $\lambda = 0.6$ in the Y-shaped bifurcation with $Q_1/Q_T = 0.75$. The flow direction and the downstream branch labels, "1" and "2" are indicated in the first photograph, Fig. 7(a). Although two spheres are visible in Fig. 7(a), the sphere in branch 1, the left branch, does not affect the motion of the sphere of interest, which is in the bifurcation. In Fig. 7(a), the sphere in the bifurcation approaches the apex of the far wall of the bifurcation. From Figs. 7(a) and 7(b), it appears that the sphere is about to enter branch 2, even though that branch receives only 25 percent of the volumetric flow. The translational velocity of the sphere decreases as the sphere nears the bifurcation wall even though its rotational velocity determined from the video recording remains appreciable. In Figs. 7(c) through 7(e), the sphere moves away from the entrance of branch 2 and is swept into branch 1 with the majority of the flow. The rotational velocity decreases once the sphere is completely within branch 1.

Figure 8 shows a similar sequence of images for a particle with $\lambda = 0.4$ in the T-shaped bifurcation for $Q_1/Q_T = 0.75$. The direction of flow is indicated in Fig. 8(a). In Figs. 8(a) through 8(c), the sphere in the bifurcation appears as though it will enter branch 2, the side branch that receives only 25 percent of the flow. However, after several seconds, the particle moves away from the side branch and into branch 1 with the majority of the flow.

Figure 9 shows another sequence of images for a sphere in the T-shaped bifurcation. In this case $Q_1/Q_T = 0.25$ and $\lambda = 0.6$. In Figs. 9(a) through 9(c), it appears that the sphere is about to enter branch 1. Instead, it is gradually swept into the

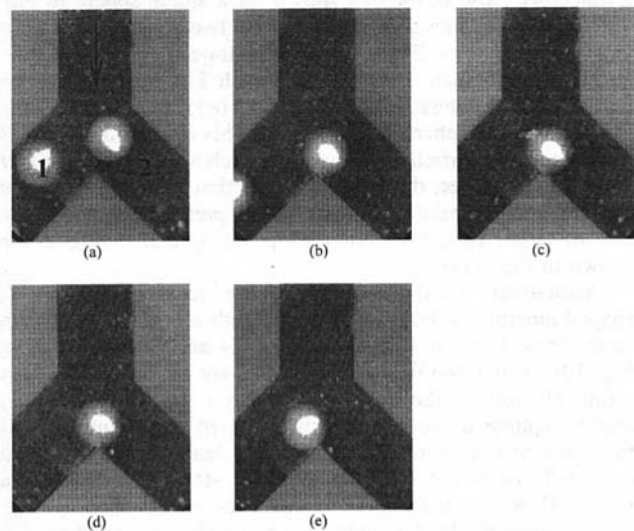


Fig. 7 A sequence of images for a sphere with $\lambda = 0.6$ in the Y-shaped bifurcation with $Q_1/Q_T = 0.75$. The time between images is 0.5 s.

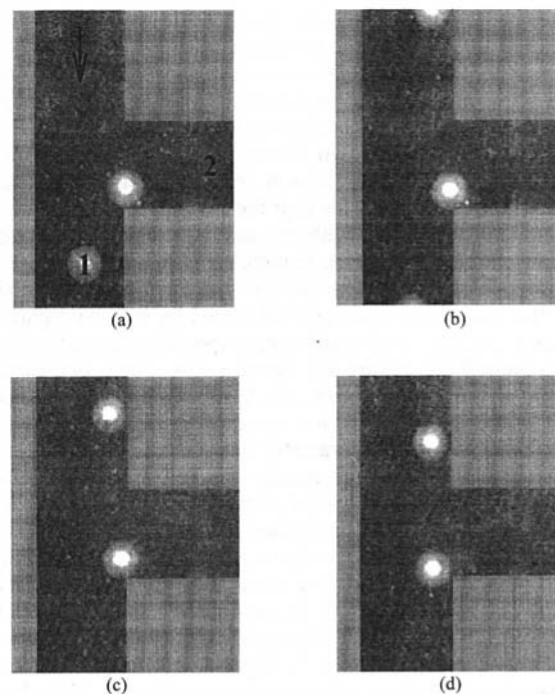


Fig. 8 A sequence of images for a sphere with $\lambda = 0.4$ in the T-shaped bifurcation for $Q_1/Q_T = 0.75$. The time between images is 0.5 s.

side branch that receives 75 percent of the flow, as shown in Figs. 9(d) and 9(e).

The motions of the spheres in Figs. 7 through 9 were selected from many observations because they illustrate the behavior of spheres that are close to the special trajectory that divides particles between the two branches. Spheres on trajectories close to the dividing trajectory move to the apex of the far wall of the bifurcation before entering one of the downstream branches. The translational velocity of a sphere is much smaller near the apex of the bifurcation in the inlet branch. Therefore, another sphere in the inlet branch can move into the bifurcation and interact with the sphere near the apex of the bifurcation. This suggests two adjacent particles are most likely to interact hydro-

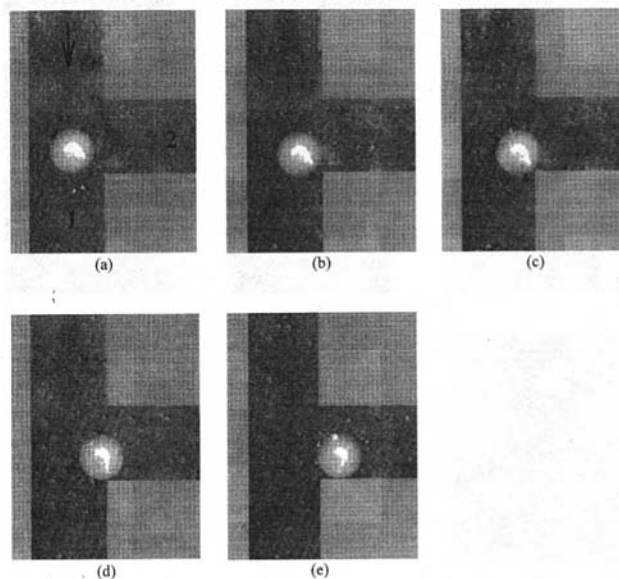


Fig. 9 A sequence of images for a sphere with $\lambda = 0.6$ in the T-shaped bifurcation for $Q_1/Q_T = 0.25$. The time between images is 0.5 s.

dynamically in the bifurcation when the downstream particle is on a trajectory close to the dividing trajectory. Particles that are not near the dividing trajectory move through the bifurcation without passing near the apex, and the velocities of these particles change only in proportion to the partitioning of the total flow rate between the two branches. The residence times of such particles in the bifurcation are small compared with the residence times of particles near the dividing trajectory, so they are unlikely to interact with the next upstream particle before moving into a downstream branch.

It remains to be determined how the trajectories of particles near the bifurcation apex are influenced by upstream particles, but this, too, can be gleaned from a careful examination of the video recordings. Figure 10 shows a case where two particles with $\lambda = 0.6$ pass through the Y-bifurcation with $Q_1/Q_T = 0.75$. In the inlet branch, the spacing between the particles is sufficiently large that interactions between the two particles are insignificant. The leading particle, marked with an "l" in the figures, enters the apex region of the bifurcation in Figs. 10(a) and 10(b) on the same trajectory it would follow if no other particles were present. However, owing to the small translational velocity of the leading particle at the apex, the trailing particle, marked with a "t" in the figures, has time to catch up to the leading particle. Eventually, the distance between the two particles becomes small, as in Fig. 10(c). As a result of the interaction, the leading sphere moves into branch 2, the branch receiving only 25 percent of the flow, rather than into branch 1, as shown in Fig. 7 for the single particle case. Although the trailing sphere moves into branch 1, the left branch, the net outcome of the interaction between the spheres is to diminish the flux of particles into branch 1, the branch receiving the greater volumetric flow rate. Because the number of interactions of this kind increases with ϕ , the importance of the effect should increase with ϕ . Indeed, the data in Figs. 3 and 4 show that

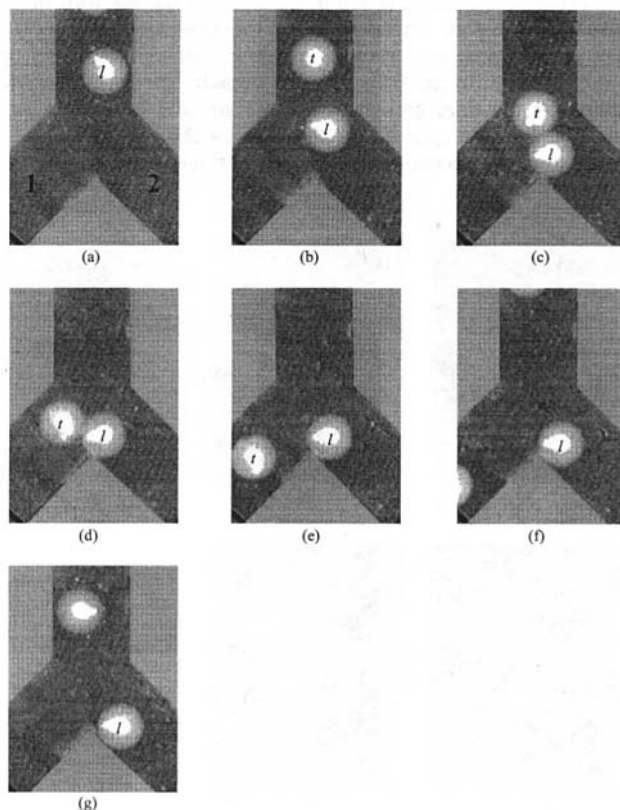


Fig. 10 A sequence of images for spheres with $\lambda = 0.6$ in the Y-shaped bifurcation $Q_1/Q_T = 0.75$. The time between images is 0.5 s.

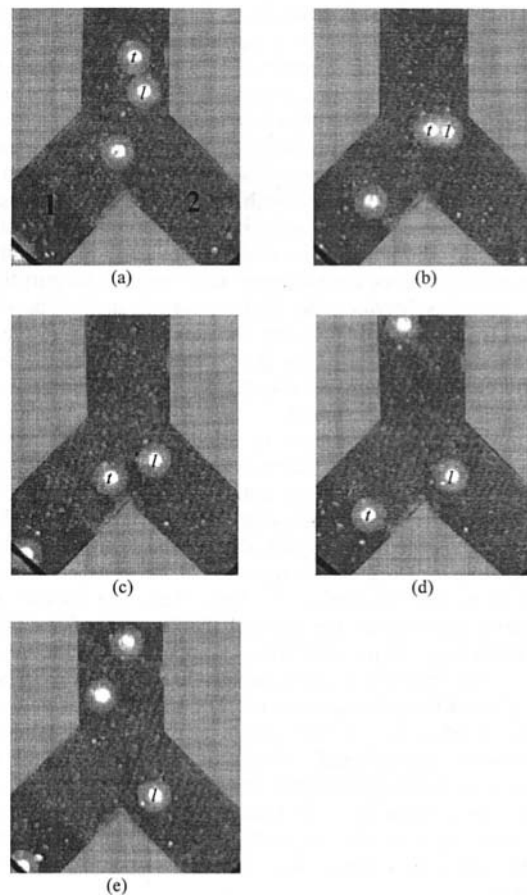


Fig. 11 A sequence of images for spheres with $\lambda = 0.4$ in the Y-shaped bifurcation with $Q_1/Q_T = 0.8$. The time between images is 0.5 s.

deviations in particle flux from the line $Q_1/Q_T = N_1/N_T$ diminish as ϕ is made larger.

A second kind of interaction is shown in Fig. 11 for particles with $\lambda = 0.4$ in the Y-shaped bifurcation with Q_1/Q_T set to 0.75. In Fig. 11(a), the spheres of interest again labeled *l* and *t*; the other sphere plays no role in the interaction. In Fig. 11(b), the two spheres interact before either one of them reaches the apex of the bifurcation. The leading sphere has a radial position in the inlet branch that would take the sphere into branch 1 on a trajectory close to the one shown for a single sphere in Fig. 7. However, the interaction between the two spheres causes the leading sphere to be displaced from that trajectory (Fig. 11(c)), and it enters branch 2 instead of branch 1 (Fig. 11(d)). The trailing sphere enters branch 1 (Fig. 11(e)), as it would have done if no other spheres were present. This interaction also acts to diminish the particle flux into the branch receiving the greater flow rate. However, this kind of interaction was only observed for the smallest particles in the study, presumably because a pair of larger particles cannot adopt the spatial configuration shown in Fig. 11(b).

Qualitatively similar interactions are observed for the T-shaped junction, at least for particles with $\lambda = 0.4$ and 0.6. An example is shown in Fig. 12 for $\lambda = 0.4$ and $Q_1/Q_T = 0.8$. In Fig. 12(a), the two upstream particles are on trajectories that ultimately would take them into branch 1. In Fig. 12(b), the leading sphere moves toward the apex of the bifurcation. If the other sphere were not present, the leading sphere would eventually be swept into branch 1, the straight branch, as in Fig. 8. However, the interaction with the other sphere in Fig. 12(c) causes the leading sphere to be displaced toward branch 2, the side branch, and it enters branch 2 in Figs. 12(d) and 12(e). An examination of video recordings shows that for fixed

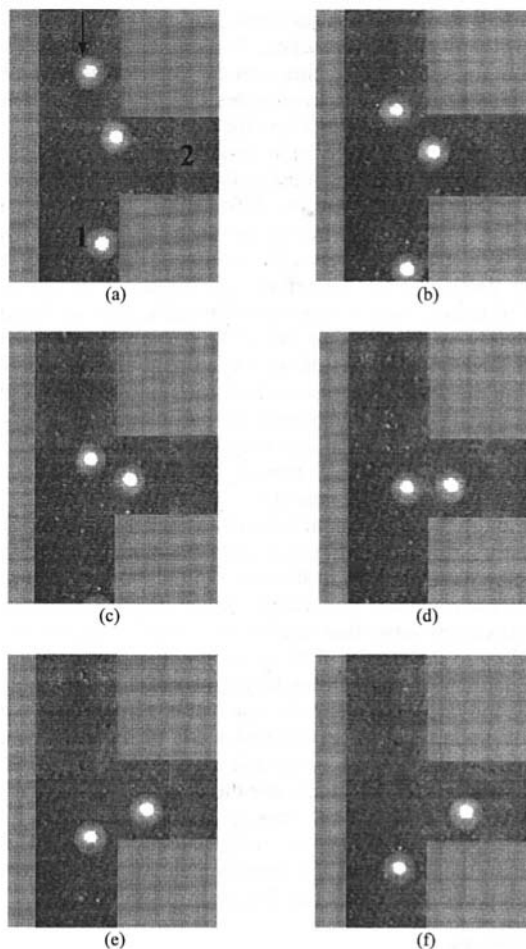


Fig. 12 A sequence of images for spheres with $\lambda = 0.4$ in the T-shaped bifurcation with $Q_1/Q_T = 0.8$. The time between images is 0.5 s.

values of λ and ϕ , binary interactions between spheres are less frequent in the T-shaped bifurcation than in the Y-shaped bifurcation. This observation is consistent with the data for particle flux. Data in Fig. 6 for the T-shaped bifurcation are less sensitive to ϕ than are the corresponding data in Fig. 4 for the Y-shaped bifurcation.

Results for the larger spheres in the T-shaped bifurcation are qualitatively different from results for the Y-shaped bifurcation when Q_1/Q_T is small. When interactions are most important, $\phi = 0.15$, N_1/N_T exceeds Q_1/Q_T for small values of Q_1/Q_T , as shown in Fig. 5. This means that the sphere flux favors the branch receiving only a small portion of the flow, which is the opposite of the result for the Y-shaped bifurcation. This seemingly odd result is a consequence of an interaction among large spheres in the T-shaped bifurcation that does not occur in the Y-shaped bifurcation. Figure 13 shows a three-particle interaction in the T-shaped bifurcation for $\lambda = 0.6$ and $Q_1/Q_T = 0.2$. Most of the flow moves into the side branch 2, but at times spheres accumulate at the entrance to the branch, as in Fig. 13(a). The two spheres at the entrance to branch 2 prevent the next upstream sphere from entering branch 2, as it would if the other spheres were not there. As a result, the third sphere moves into branch 1 in Figs. 13(b) and 13(c), even though branch 1 receives only 20 percent of the flow. This kind of interaction was only observed for $\phi = 0.15$, which is not surprising, because it requires three particles in the bifurcation simultaneously. (One particle could, in principle, block the entrance to the side branch, but this happens only rarely.) Although this kind of interaction is relatively infrequent, it occurs often enough to increase the flux into branch 1 for small Q_1/Q_T , as

shown in Fig. 5 for $\phi = 0.15$. A similar result is seen in figure 6, but the effect is smaller because blocking is infrequent for smaller particles.

For larger values of Q_1/Q_T ($Q_1/Q_T > 0.6$), the most striking difference between results for the two bifurcations occurs for the largest particles $\lambda = 0.8$. The results in Fig. 5 show that the particle flux is practically independent of ϕ for the T-shaped bifurcation, even though it depends strongly on ϕ for the Y-shaped bifurcation, cf. Fig. 3. This suggests that there is some interaction in the Y-shaped bifurcation that does not occur in the T-shaped bifurcation. The interaction that is "missing" in the T-shaped bifurcation is the one illustrated in Fig. 10 for the Y-shaped bifurcation. Figure 10 showed that a sphere at the bifurcation apex can be diverted by interactions with another sphere into the opposite branch than the one it enters in the absence of interactions. For this to occur, the first sphere must reside at the apex for a sufficient time to encounter the second sphere. This interaction simply does not occur for large spheres in the T-shaped bifurcation. The videorecordings show that the residence time of a leading sphere near the apex of the T-shaped bifurcation is less than the residence time of a sphere near the apex of the Y-shaped bifurcation. In the T-shaped bifurcation the residence time of the leading sphere at the apex is insufficient for an interaction with another sphere to change its path. Thus, the mechanism by which interactions change the particle flux in the Y-shaped bifurcation is not observed in the T-shaped bifurcation for large Q_1/Q_T , and the dependence of N_1/N_T on ϕ vanishes in Fig. 5. For smaller particles in the T-shaped bifurcation, this interaction mechanism is recovered because the residence time of the smaller particles near the apex can be large. Indeed, the particle flux depends on ϕ , albeit weakly, for smaller λ in Fig. 6.

Comparisons With Previous Studies. The models experiments of Chien et al. [4] are closely related to the present work. Chien et al. studied the motion of rigid spheres and flexible disks through a symmetric T-bifurcation, which corresponds to $\theta_1 = \theta_2 = \pi/2$ in Fig. 1 (top). They studied suspensions of spheres with values of 1 from 0.32 to 0.79, but for only one particle volume fraction, $\phi = 0.05$.

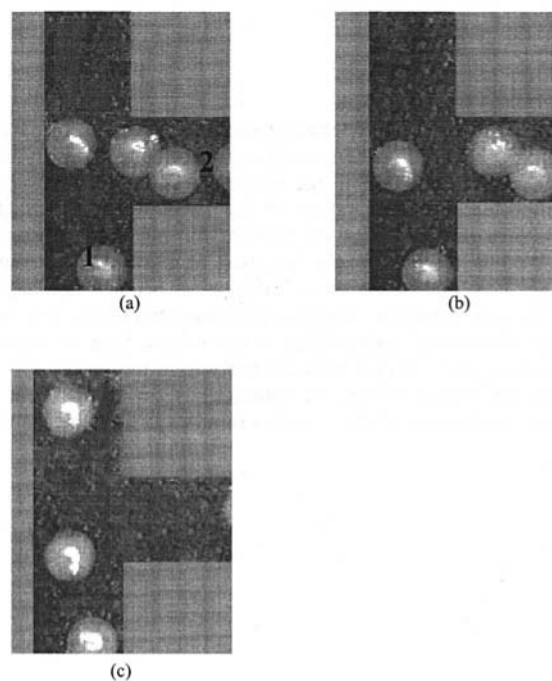


Fig. 13 A sequence of images for spheres with $\lambda = 0.6$ in the T-shaped bifurcation with $Q_1/Q_T = 0.2$. The time between images is 0.5 s.

Their results for $\lambda = 0.32$ agree closely with results of this experiment for $\lambda = 0.4$. In both experiments, data fall on the line $N_1/N_T = Q_1/Q_T$ except when Q_1/Q_T is near zero or near unity. Then, there is a measurable difference between N_1/N_T and Q_1/Q_T as in Fig. 4 for the Y-shaped bifurcation. Comparisons for the case $\lambda = 0.48$ are ambiguous because there are few data in [4] for values of Q_1/Q_T near 0.5. For larger values of λ , the data of the present study are similar to those in [4], although there are numerical differences, which are expected since the values of ϕ and the shapes of the bifurcation differ in the two experiments.

Yen and Fung [10] studied the flux of flexible disks through a symmetric T-bifurcation for values of ϕ up to 0.30. The effect of ϕ in their study is similar to the effect observed here, viz., that the difference between N_1/N_T and Q_1/Q_T for a given value of Q_1/Q_T is diminished as ϕ is made larger. The magnitude of the effect is smaller in [10] than in the data in Fig. 3, which may reflect the shape of the bifurcation and the shape and deformability of the particles in [10], although it is impossible to be certain from the available data.

Cell flux at bifurcations often is interpreted, at least qualitatively, in terms of plasma skimming, which reflects nonuniform cell profiles in the inlet vessel, and cell screening, which reflects hydrodynamic interactions between the particles and the channel walls. Enden and Popel [6] developed a numerical model of plasma skimming in an asymmetric T-shaped bifurcation similar to the one in this study. Blood is treated as a homogeneous fluid in the model, which eliminates any effect of cell screening on the particle flux. The model shows that the effect of plasma skimming on cell partitioning is independent of the branch angle and is unaffected by the asymmetry of the bifurcation. In the present experiment, the particle flux depends strongly on the shape of the bifurcation and on the sphere size λ , both of which suggest the importance of cell screening on the data in this study.

Data from the in vivo study of 65 bifurcations by Pries et al. [8] also show no effect of branching angle or of asymmetry of the bifurcation geometry on cell partitioning. However, these results rely on lumped data that span a wide range of capillary diameters. It is possible that the lumped data are dominated by effects of plasma skimming, while cell screening is important only for the subset of bifurcations composed of smaller diameter vessels.

Conclusions

The results of this experiment show that the flux of particles through a bifurcation can be strongly influenced by hydrodynamic interactions among the particles. These interactions can be important at small values of the particle volume fraction, because of the nature of the flow at the bifurcation. Particles that are spaced sufficiently far apart as they move through a straight tube can interact in a bifurcation because the velocity of a particle near the dividing trajectory becomes very small in the bifurcation, and trailing spheres have time to enter the bifurcation and interact with the particle at the apex.

The qualitative effects of interactions on the particle flux depend on the size of the particles relative to the tube diameter,

the overall shape of the bifurcation, and the partitioning of total volume between the branches. The influence of interactions on data for the particle flux can be correlated with detailed examinations of particle trajectories that lead to the identification of several basic interaction mechanisms.

The particle flux data for a suspension that contains many particles represent a statistical average of all possible interactions among particles at the bifurcation. In the absence of Brownian motion and in the absence of nonhydrodynamic forces between the particles, the motion of the particles is completely deterministic. Therefore, the frequency of the various kinds of interactions in a given bifurcation flow is determined completely by the arrangement of the particles in the inlet tube. In this sense, the results of the experiment can be regarded as showing the relative importance of each interaction on particle flux for a random arrangement of particles in the inlet tube. Other interaction mechanisms may be possible if the configuration of particles in the inlet branch, such as their radial positions or axial spacing, is not random.

Although the experiment is hardly a realistic model of microcirculatory flow, the effects of particle interactions displayed here may have analogs in the microcirculation. Hematocrits in the microcirculation are usually lower than systemic values, but sufficiently large that interactions could influence cell flux, according to the results of the present experiment. Microscopic examinations of microcirculatory flow often observe relatively slowly moving cells near the apex of a bifurcation, which is the key hydrodynamic effect that leads to interactions among cells at the bifurcation. Although the present experiment has neither the correct cell shape nor the deformability of real cells, which certainly affects cell motion through the bifurcation, the results suggest the importance of hematocrit, the detailed shape of the bifurcation, and the arrangement of cells in the inlet vessel on cell partitioning at divergent bifurcations.

References

- 1 Audet, D. M., and Olbricht, W. L., "The Motion of Model Cells at Capillary Bifurcations," *Microvascular Research*, Vol. 33, 1987, pp. 377–396.
- 2 Bugliarello, G., and Hsiao, G. C. C., "Phase Separation in Suspensions Flowing through Bifurcations: A Simplified Hemodynamic Model," *Science*, Vol. 143, 1964, pp. 469–471.
- 3 Carr, R. T., and Wickham, L. L., "Influence of Vessel Diameter on Red Cell Distribution at Microvascular Bifurcations," *Microvascular Research*, Vol. 41, 1991, pp. 184–196.
- 4 Chien, S., Tvetenstrand, C. D., Farrell Epstein, M. S., and Schmid-Schonbein, G. W., "Model Studies on Distributions of Blood Cells at Microvascular Bifurcations," *American Journal of Physiology*, Vol. 248, 1985, pp. H568–H576.
- 5 Dellimore, J. W., Dunlop, M. J., and Canham, P. B., "Ratio of Cells and Plasma in Blood Flowing Past Branches in Small Plastic Channels," *American Journal of Physiology*, Vol. 244, 1983, pp. H635–H643.
- 6 Enden, G., and Popel, A. S., "A Numerical Study of Plasma Skimming in Small Vascular Bifurcations," *ASME JOURNAL OF BIOMECHANICAL ENGINEERING*, Vol. 116, 1994, pp. 79–88.
- 7 Fung, Y.-C., "Stochastic Flow in Capillary Blood Vessels," *Microvascular Research*, Vol. 5, 1973, pp. 34–48.
- 8 Pries, A. R., Ley, K., Claassen, M., and Gaetgens, P., "Red Cell Distribution at Microvascular Bifurcations," *Microvascular Research*, Vol. 38, 1989, pp. 81–101.
- 9 Yan, Z.-Y., Weinbaum, S., and Acrivos, A., "A Three-dimensional Analysis of Plasma Skimming at Microvascular Bifurcations," *Microvascular Research*, Vol. 42, 1991, pp. 17–38.
- 10 Yen, R. T., and Fung, Y. C., "Effect of Velocity Distribution on Red Cell Distribution in Capillary Blood Vessels," *American Journal of Physiology*, Vol. 235, 1978, pp. H251–H257.